Lab 8 – Frequency Response*
EE252L/251L: Signals and Systems Lab

Dr. Basit Memon and Ms. Hira Mustafa

Name: _____

Objectives

By the end of this lab, you will be able to get familiarize with the designing and analysis of frequency responses of FIR filters on MATLAB.

Instructions: After completion of each task get it marked by sharing your results and code with any of the Instructor, your code must be self-explanatory i.e. use proper comments. *Attachrubrics at the end of your report*.

1 Frequency Response

In signal processing, it is often helpful to employ alternate representations of a signal in order to emphasize and extract relevant information from the signal. In signal processing applications today, a number of signal representations are in practice, including Frequency, Fourier, Cosine Transforms, Wavelet decomposition, Eigenvectors, Principal Components, and many more. In today's lab, we'll focus on determining the frequency representations of signals and system, specifically the frequency response of FIR filters.

The frequency response of a system, denoted by $H(j\omega)$, characterizes the output of the system to input signals at various frequencies. There are infinite possibilities for input signals, so how could we characterize the output of a system to such a vast range of inputs and generate this frequency response? The task is doable for LTI systems, because of two properties: the output of the system to an input sinusoidal signal at any frequency is the sinusoidal signal with the same frequency, but possibly different amplitude and phase; and any input signal can be represented as a sum of sinusoids (more on this later) and so the output of an LTI system to any input signal can be determined using linearity. So, the frequency response of an LTI system is simply a characterization of the dependence of the amplitude and the phase of output of the system in

*Adapted from DSP First, 2e

response to sinusoidal inputs of various frequencies.

The frequency response of an FIR filter will inform us of how the filter will affect the amplitude and phase of an input signal at a specific frequency. For example, the frequency response of the two-point averaging filter

$$y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$$

can be found using a general complex exponential, $x[n] = Ae^{j(\omega n+\phi)}$, and observing the output.

$$\begin{aligned} x[n] &= Ae^{j(\omega n+\phi)} \\ \Rightarrow y[n] &= \frac{A}{2}e^{j(\omega n+\phi)} + \frac{A}{2}e^{j(\omega(n-1)+\phi)} \\ &= Ae^{j(\omega n+\phi)}\left(\frac{1}{2} + \frac{1}{2}e^{-j\omega}\right) \\ &= x[n]\underbrace{\left(\frac{1}{2} + \frac{1}{2}e^{-j\omega}\right)}_{H(j\omega)} \end{aligned}$$

Notice that output is a product of two terms - the input and a term that is a function of ω only. This second term is the frequency response of the filter.

1.1 MATLAB function for Frequency Response

We can use MATLAB's `freqz` function to find the frequency response of a filter. As an example, the following code computes and plots both the magnitude and phase of the frequency response of the two-point averaging filter:

```
bb = [0.5, 0.5]; %-- Filter Coefficients
ww = -pi:(pi/100):pi; %-- Frequency vector
H = freqz(bb, 1, ww); %frequency reponse
subplot(2,1,1);
plot(ww, abs(H)), grid on
subplot(2,1,2);
plot(ww, angle(H)), grid on
xlabel(Normalized Radian Frequency)
```

Notes:

1. For FIR filters, the second argument of `freqz` will always be equal to 1.
2. It is enough to plot the frequency response of discrete-time filters over a frequency range of 2π as they are always periodic with period 2π. If you get a chance, think about why is that true.

Task 1: Frequency Response of K-point moving average filter

We have used a K-point moving average filter in the previous labs and we know that it is described by:

$$y[n] = \frac{1}{K}\sum_{l=0}^{K-1} x[n-l].$$

(a) Find the frequency response of a four-point averaging filter, i.e.:

$$y[n] = \frac{1}{4}\sum_{l=0}^{3} x[n-l], 0 \leq n<4,$$

using MATLAB's `freqz` function and plot its magnitude and phase versus ω.
Hint: Use `ones` to create filter coefficient matrix.

(b) Looking at the frequency response, argue about the reason that this filter had a smoothing effect on the signals in the previous labs.

(c) Use Euler's formula and complex number manipulations to show that the frequency response for the 4-point moving average filter is:

$$H(j\omega) = \frac{2\cos(0.5\omega) + 2\cos(1.5\omega)}{4} e^{-j1.5\omega}.$$

Implement the expression for $H(j\omega)$ in MATLAB by choosing 400 samples between $-\pi$ and π for ω. Using `abs` and `angle`, extract the magnitude and phase of this frequency response and plot it against frequency. Compare your results to the plots obtained in the previous part.

(d) Implement a K-point averaging filter in MATLAB and observe the effect of varying K, the filter length, on its frequency response. State your observations and comment on them.

2 Band-pass Filter

A band pass filter allows frequencies within a specific frequency range and rejects (attenuates) frequencies outside that range. We can construct a band-pass filter by using an L-point FIR with impulse response:

$$h[n] = \frac{2}{L} cos(\omega_c n), 0 \leq n < L$$

where L is the length of the filter and ω_c is the center frequency for the pass-band. The passband of the band-pass filter is the region of allowable frequencies.

Task 2: Band pass filter

(a) Plot the frequency response of this band-pass filter for $\omega_c = 0.44\pi$ and $L = 10$. You can again use the `freqz` function. The values of the impulse response are the coefficients of this filter.

(b) Plot the frequency response for $L = 20$ and $L = 40$.
Now, If we define the maximum value of $| H(j\omega) |$ as H_{max} then conventionally the passband width is defined as the length of the frequency region where the ratio $| H(j\omega) | / H_{max}$ is greater than $1/\sqrt{2} = 0.707$.

- Using the above relation determine the width of the previous filters.
  (Hint: MATLAB's `find` command can be used.)
- What is the effect of doubling L on the passband of the filter?

(c) Design a band-pass filter that will pass the frequency component at $\omega_c = 0.44\pi$ but reduce the components near $\omega = 0.3\pi$ and $\omega = 0.7\pi$. The smallest value of L should be chosen for which any frequency component satisfying $|\omega| \leq 0.3\pi$ or $0.7\pi \leq |\omega| \leq \pi$ will be reduced by a factor of 10 or more, i.e. $|H(j\omega)| \leq \frac{H_{max}}{10}$.

(d) Test your filter by applying it to the signal:

$$x[n] = 10cos(0.3\pi n) + 40cos(0.44\pi n - \pi/3) + 20cos(0.7\pi n - \pi/4).$$

- First find the output y[n].
- Now after finding the output we are going to find the frequency representation of the output signal using `fft`[a].

```
fs = 2*pi;    % sampling frequency
t = 0:(1/fs):(10-1/fs);  % time vector
 S= y;    %output acquired in step 1
n = length(S);
X = fft(S) ;
Y = fftshift(X);
fshift = (-n/2:n/2-1)*(fs/n); % zero-centered frequency range
plot(fshift,abs(Y)/n)
```

- Observe the response and comment on the filter you have designed.
  Note:In frequency response graph the x-axis represents frequency in Hz.

[a] You'll become intimately familiar with this in a future lab.

Assessment Rubric

Lab 8: Frequency Response

Name:	**Student ID:**

Marks distribution:

	Tasks	**LR2 Code**	**LR5 Results**	**LR9 Report**	**AR4 Report Submission**	**AR7 Comments**
In-Lab	**Task 1**	/8	/12	/8	/12	
	Task 2	/16	/8	/8		/8
Total =80		/24	/20	/16	/12	/8
CLO Mapped		**CLO2**	**CLO2**	**CLO2**	**CLO3**	**CLO3**

CLO – (LDL)	**Total Points**	**Points Obtained**
CLO2 – (Psy-4)	60	
CLO3 – (Aff-3)	20	
	80	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	**Data Collection**	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	**Calculations**	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR11	**Design**	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR4	***Report Submission**	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.
AR7	**Report Content/Code comments**	Most of the questions are not answered / figures are not labelled/ titles are not mentioned / units are not mentioned. No comments are present in the code.	Some of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Few comments are stated in the code.	Majority of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Comments are stated in the code.	All the questions are answered, figures are labelled, titles are mentioned and units are properly mentioned. Proper comments are stated in the code.